Vector Calculus and Complex Analysis Midterm II Solution

1. Find all values of $(1+i)^{1-i}$.

Solution

$(1+i)^{1-i} = e^{(1-i)\ln(1+i)}$ (1)

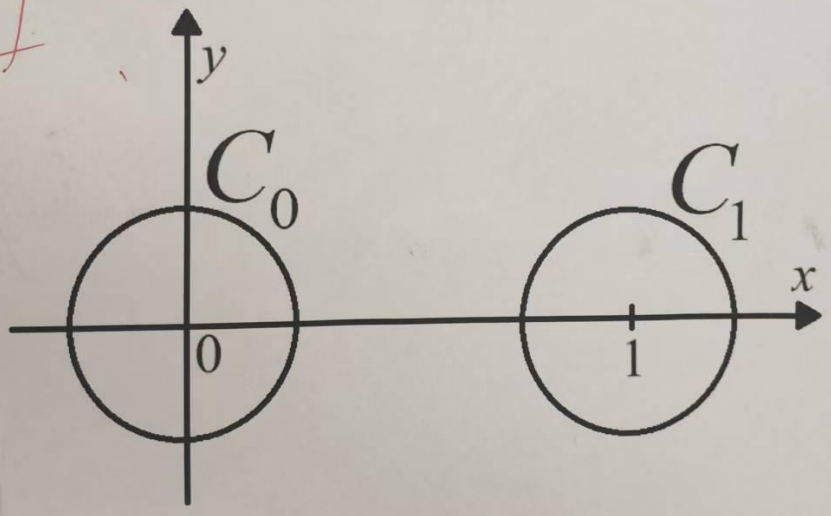
$1+i = \sqrt{2}e^{i(\pi/4+2n\pi)}$

$\ln(1+i) = \frac{1}{2}\mathrm{Log}2 + i(\frac{\pi}{4}+2n\pi)$ (1)

$(1-i)\ln(1+i) = (1-i)\frac{1}{2}\mathrm{Log}2 + (1-i)i(\frac{\pi}{4}+2n\pi) = \frac{1}{2}\mathrm{Log}2 + \frac{\pi}{4} + 2n\pi - i(\frac{1}{2}\mathrm{Log}2 + \frac{\pi}{4} + 2n\pi)$ (2)

$(1+i)^{1-i} = e^{(1-i)\ln(1+i)} = e^{\frac{1}{2}\mathrm{Log}2+\frac{\pi}{4}+2n\pi-i(\frac{1}{2}\mathrm{Log}2+\frac{\pi}{4}+2n\pi)}$ (1)

2. Consider the following curves.



Suppose $\int_{C_1} \frac{z^2+bz+c}{z^2(z-1)}dz = 2\pi$. Calculate $\int_{C_0} \frac{z^2+bz+c}{z^2(z-1)}dz$.

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Solution

$$2\pi = \int_{C_1} \frac{z^2+bz+c}{z^2(z-1)}\,dz = \int_{C_1} \frac{(z^2+bz+c)/z^2}{(z-1)}\,dz = 2\pi i(1+b+c) \quad (1)$$

$$\int_{C_0} \frac{z^2+bz+c}{z^2(z-1)}\,dz = \int_{C_0} \frac{(z^2+bz+c)/(z-1)}{z^2}\,dz$$

$$= 2\pi i \frac{d}{dz}\left(\frac{z^2+bz+c}{z-1}\right)\Bigg|_{z=0}$$

$$= 2\pi i \left(\frac{(z-1)(2z+b)-(z^2+bz+c)}{(z-1)^2}\right)\Bigg|_{z=0} \quad (1)'$$

$$= 2\pi i(-b-c) \quad (1)$$

$$= 2\pi i - 2\pi \quad (1)$$

$1 = i(1+b+c)$.

$i(b+c) + i$

$-i(b+c) = i \cdot 1$

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3. (a) Show that $\mathrm{Re}\left(\frac{z+1}{z-1}\right) < 0$ if $|z| < 1$.

(b) Let $f(z) = e^{\frac{z+1}{z-1}}$. Show that $|f(z)| \le 1$ for all $|z| < 1$ and $|f^{(n)}(0)| \le n!$

Solution

$$\frac{z+1}{z-1} = \frac{(z+1)(\bar{z}-1)}{|z-1|^2} = \frac{|z|^2-1-z+\bar{z}}{|z-1|^2} = \frac{|z|^2-1-2iy}{|z-1|^2} \quad (1)$$

$$\mathrm{Re}\left(\frac{z+1}{z-1}\right) = \frac{|z|^2-1}{|z-1|^2} < 0 \quad \text{if } |z| < 1$$

$$\left|e^{\frac{z+1}{z-1}}\right| = \left|e^{u+iv}\right| = e^u < 1 \quad \text{if } |z| < 1 \quad (1)$$

$$\left|f^{(n)}(0)\right| \le \frac{n!}{R^n} \quad \text{for any } R < 1$$

$$\le n! \quad \text{as } R \to 1^- \quad (1)'$$

$f^n(z_0) = \frac{n!}{2\pi i}\oint_C \frac{f(z)}{(z-z_0)^{n+1}}\,dz$.